CAR/AUTO REVERSE PARKING SECURITY SYSTEM With Audio Video Alert

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Fig. 1: Author's prototype

Parking safely in a garage or other confined space can be challenging, particularly when rear visibility is limited. Even a slight miscalculation while reversing can damage the vehicle, nearby walls, pillars, or other parked vehicles. Although many modern cars come with factory-fitted parking sensors, retrofitting them to older vehicles can be expensive. This system can also be installed in auto-rickshaws and other vehicles, providing an audible warning and displaying STOP to alert the driver before a collision occurs. Fig. 1 shows the author's prototype.

The system is based on an Arduino Nano and an HC-SR04 ultrasonic distance sensor. The sensor continuously measures the distance between the vehicle and the nearest obstacle by transmitting high-frequency sound waves and detecting their reflected echoes. The Arduino Nano processes the measured distance and compares it with a preset safety limit of 100cm. When an obstacle comes within this range, the controller activates a piezo buzzer through a BC547 transistor and displays STOP on a four-digit 7-segment LED display, providing both audible and visual warnings. The circuit operates off a regulated 12V DC supply, with BC547 transistors used to drive the display digits. Fig. 2 shows the complete system and its working principle.

Circuit and working

The system is built around an Arduino Nano (BOARD1), an HC-SR04 ultrasonic sensor, four common-cathode 7-segment LED displays, five BC547 transistors, a buzzer, and a few other components.

The circuit is powered by a regulated 12V DC supply connected to the Arduino Nano. The Arduino Nano provides the control logic, while the HC-SR04 ultrasonic sensor continuously measures the distance between the vehicle and the nearest obstacle. The output pins of the Arduino Nano are connected to the bases of five BC547 transistors. Four transistors drive the multiplexed 7-segment LED displays, while the fifth transistor drives the buzzer. The HC-SR04 ultrasonic sensor has four terminals: VCC, GND, TRIG, and ECHO. During operation, the Arduino sends a 10µs pulse to the TRIG pin, causing the sensor to transmit an ultrasonic burst at 40kHz.

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Ready to bring this circuit to life? To unlock the circuit diagram, component list, PCB layout and complete step-by-step build instructions visit: https://efy.co.in/oiw133